

EXPERIMENT 16

Implement Feed Forward Neural Network

Aim

To implement a **Feed Forward Neural Network (FFNN)** using Python for binary classification.

Algorithm

1. Initialize the input data and target output.
2. Initialize weights and biases.
3. Perform **forward propagation** from input layer to hidden layer and then to output layer.
4. Calculate the predicted output.
5. Calculate the error between actual and predicted output.
6. Update the weights and biases using the error.
7. Repeat the process for several epochs.
8. Display the final predicted output.

Program

```
import math

# Sigmoid activation function
def sigmoid(x):
    return 1 / (1 + math.exp(-x))

# Training data: OR gate
X = [
    [0, 0],
    [0, 1],
    [1, 0],
```

```
[1, 1]  
]
```

```
Y = [0, 1, 1, 1]
```

```
# Weights
```

```
w1 = 0.5
```

```
w2 = 0.5
```

```
bias = -0.2
```

```
learning_rate = 0.5
```

```
# Training
```

```
for epoch in range(1000):
```

```
    for i in range(len(X)):
```

```
        # Forward propagation
```

```
        z = X[i][0] * w1 + X[i][1] * w2 + bias
```

```
        output = sigmoid(z)
```

```
        # Error
```

```
        error = Y[i] - output
```

```
        # Weight update
```

```
        w1 = w1 + learning_rate * error * X[i][0]
```

```
        w2 = w2 + learning_rate * error * X[i][1]
```

```
        bias = bias + learning_rate * error
```

```
# Testing

print("Feed Forward Neural Network Output:")

for i in range(len(X)):

    z = X[i][0] * w1 + X[i][1] * w2 + bias
    output = sigmoid(z)

    if output >= 0.5:
        prediction = 1
    else:
        prediction = 0

    print(X[i], "->", prediction)
```

Input

[0, 0]

[0, 1]

[1, 0]

[1, 1]

Output

```
Feed Forward Neural Network Output:
[0, 0] -> 0
[0, 1] -> 1
[1, 0] -> 1
[1, 1] -> 1
```

Result

Thus, the Feed Forward Neural Network was successfully implemented in Python and correctly learned the OR gate classification.

